Machine Learning Praxis

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Preface

> "One man's ways may be as good as another's, but we all like our own best." Jane Austen, *Persuasion* (1816).

A series of scripts introducing basic machine learning, natural language processing, and large language models using the python ecosystem of packages.

1 Introduction

This is a book created from markdown and executable code.

See (Knuth 1984) for additional discussion of literate programming.

Knuth, Donald E. 1984. "Literate Programming." *Comput. J.* (USA) 27 (2): 97–111. https://doi.org/10.1093/comjnl/27.2.97.

2 EDA

EDA placeholder

3 Classification

Classification placeholder

4 Regression

Regression placeholder

5 Clustering

Clustering placeholder

6 NLP

Natural Language Processing (NLP) bridges the gap between text and computational models, enabling algorithms to parse, understand, generate, and extract meaningful patterns from unstructured text data.

6.1 (Pre)processing Text

Core steps in preparing raw text for natural language processing (NLP) analysis:

1. **Lowercasing**: Convert all characters in the text to lowercase to ensure that words with different capitalizations (e.g., "Data", "DATA", and "data") are treated as identical tokens by models.
2. **Tokenization**: Split the continuous stream of text into smaller, discrete units, usually individual words, subwords, or sentences, using libraries like NLTK or spaCy.
3. **Removing Noise and Special Characters**: Strip out irrelevant elements such as HTML tags, URLs, emojis, and punctuation marks that do not add meaningful semantic value to the analysis.
4. **Removing Stop Words**: Filter out extremely common, high-frequency words (such as "the", "is", "at", and "and") that carry little to no substantive thematic weight.
5. **Normalization (Stemming & Lemmatization)**: Reduce inflected or derived words down to their root base forms. *Stemming* uses crude heuristic rules to chop off word suffixes, while *lemmatization* uses vocabulary and morphological analysis to return the proper dictionary form (the lemma).

6. **Vectorization**: Convert the cleaned, tokenized text into numerical representations (such as TF-IDF matrices, word embeddings, or transformer-based token IDs) so that machine learning algorithms and language models can process them mathematically.

6.2 Text Summarization

Text summarization is a natural language processing task that uses machine learning to distill a long document into a shorter version while preserving its core meaning, key information, and overall narrative.

6.3 Topic Modeling

Topic modeling is an unsupervised machine learning technique that automatically scans documents to cluster words and uncover the hidden semantic themes or topics running through a large corpus of text.

6.4 Keywords-In-Context (KWIC)

Keywords in context (KWIC) is a text analysis technique that displays a targeted search term alongside its immediate surrounding words to help quickly examine semantic usage patterns, sentiment, and linguistic context across a large corpus.

7 LLM

LLM placeholder

8 Summary

In summary, this book has no content whatsoever.